

**PHALANX XDR: AN EXTENDED DETECTION AND RESPONSE
PLATFORM
PROJECT REPORT**

Submitted

BY

MBAZIIRA JOEL JIMMY

Roll No.: 012221086

In Partial Fullfillment of the Requirements for the Award of the Degree of
BACHELOR OF SCIENCE IN COMPUTER ENGINEERING

**Under the Guidance
of
DR.MAYUR KUMAR**



**ISBAT
UNIVERSITY**
A CHARTERED UNIVERSITY

FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

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Starting Year: 2022 Graduating Year: 2026

_____ to _____, _____
(Start Month) (End Month) (Year)

DECLARATION

I the undersigned solemnly declare that the report of the project work entitled **Phalanx XDR: An Extended Detection and Response Platform**, is based my own work carried out during the course of my study under the supervision of **Dr. Mayur Kumar** as requirements of six months project semester for the award of degree of **BS.CE. (Bachelor of Science in Computer Engineering)**, ISBAT University, Kampala. I assert that the statements made and conclusions drawn are an outcome of the project work. I further declare that to the best of my knowledge and belief that the project report does not contain any part of any work which has been submitted for the award of any other degree/diploma/certificate in this University or any other University.

(Signature)
MBAZIIRA JOEL JIMMY
012221086

Date: _____

Certified that the above statement made by the student is correct to the best of our knowledge and belief.

DR MAYUR KUMAR
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ABSTRACT

Modern enterprise environments face an escalating volume of sophisticated cyber threats that bypass traditional perimeter defenses. While Extended Detection and Response (XDR) architectures offer cross-layered visibility across networks and endpoints, commercial solutions remain financially prohibitive and operationally complex for small-to-medium enterprises (SMEs). This project presents Phalanx XDR, a comprehensive, unified, open-source XDR platform engineered to bridge this security gap.

The system architecture features a dual-layer data collection framework: a Python-based network sensor (`phalanx_sensor`) and a cross-platform Python endpoint agent (`sentinel-py`). The `phalanx_sensor` node provides network visibility by executing real-time DHCP watching, passive OS fingerprinting via OUI/TTL analysis, and DNS sniffer protocols to detect Domain Generation Algorithms (DGA) and network beaconing. Concurrently, the `sentinel-py` agent gathers telemetry across six host domains: process lifecycle, file integrity, network sockets, DNS queries, user logins, and memory anomalies.

The backend processing engine, engineered on the .NET 8 framework, decouples data ingestion from complex event evaluation. Telemetry is received via a high-throughput

REST API (`Phalanx.Ingestion`) and buffered directly into an in-memory Redis cache pipeline. A dedicated background service (`Phalanx.Correlator`) asynchronously consumes these events, evaluating stateful and sequence-based correlation rules written in a flexible YAML schema mapped precisely to the MITRE ATT&CK enterprise matrix. Upon detecting malicious behavior, the automated response playbooks dispatch targeted containment commands leveraging Linux `nftables`, traffic shaping (`tc`), and `dnsmasq` sinkholing on the network sensor, alongside process termination on endpoints. Analysts interact with the platform through an ASP.NET MVC web interface (`Phalanx.Console`) featuring real-time alerting powered by SignalR. Experimental validation confirms that Phalanx XDR delivers ultra-low latency ingestion, highly accurate

multi-stage attack correlation, and near-instantaneous host isolation, establishing a viable, robust open-source alternative for enterprise defense.

Keywords: Extended Detection and Response (XDR), Telemetry Ingestion, Stateful Event Correlation, Linux Network Isolation, Endpoint Monitoring, In-Memory Caching.

Computing Review Subject Codes (ACM CCS 2012):

- Security and privacy ~ Network security ~ Intrusion detection systems
- Security and privacy ~ Systems security ~ Software security engineering
- Information systems ~ Information storage systems ~ Record management systems
- Software and its engineering ~ Software organization and properties ~ Contextual software domains

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Chapter 1: Introduction

1.1 Background of the Study

The contemporary enterprise information technology landscape is defined by pervasive connectivity, cloud adoption, and a highly decentralized workforce. While these advancements drive operational efficiency, they simultaneously expand the corporate attack surface, exposing institutions to sophisticated cyber-adversaries. Traditional defensive frameworks rely heavily on siloed, perimeter-centric security solutions, such as standalone firewalls, signature-based Intrusion Detection Systems (IDS), and traditional Endpoint Protection Platforms (EPP). However, modern multi-stage attack campaigns such as advanced persistent threats (APTs), ransomware deployments, and stealthy lateral movements frequently bypass isolated defenses by operating within the blind spots (Akre et al., 2025; Albasheer et al., 2022) left between disconnected telemetry streams. To counteract these unified tactics, the cybersecurity industry evolved toward Endpoint Detection and Response (EDR) and subsequently Extended Detection and Response (XDR) systems (Akre et al., 2025; Albasheer et al., 2022). XDR coordinates security telemetry from traditionally disparate vectors including endpoints, local network traffic, DNS logs, and user authentication schemas into a centralized processing hub. By correlating multi-vector events in real-time, XDR enables security operations centers (SOC) to uncover complex malicious behavioral patterns that would otherwise appear benign when viewed in isolation. Despite the proven efficacy of XDR architectures, mainstream enterprise adoption remains deeply constrained by structural economic barriers. Leading commercial XDR platforms utilize aggressive, volume-based ingestion pricing or restrictive per-seat licensing models. For small-to-medium enterprises (SMEs) and educational infrastructures within developing regions (Bassey et al., 2024), these costs are financially unfeasible. Furthermore, existing open-source alternatives, such as standalone Security Information and Event Management (SIEM) systems, frequently suffer from extreme operational fragmentation. They require

complicated, manual integration (Bassey et al., 2024) across independent ingestion pipelines, open-source agents, and visualizers. Consequently, there is a critical requirement (Bassey et al., 2024; Albasheer et al., 2022) for an accessible, consolidated, and cohesive open-source XDR solution capable of orchestrating both active network sniffing and endpoint monitoring beneath a unified detection and response fabric.

1.2 Problem Statement

Enterprises are increasingly vulnerable to multi-vector cyberattacks because legacy security architectures process network traffic and endpoint events in complete isolation.

When an adversary executes a stealthy attack chain—such as initiating an initial phishing foothold, performing OS fingerprinting, scanning for internal assets, establishing command-and-control (C2) channels via Domain Generation Algorithms (DGA), and ultimately dumping credentials the technical indicators are fragmented across disparate logs. Standard monitoring tools fail to link these individual occurrences into a single, cohesive incident narrative.

While commercial XDR suites solve this problem by providing unified correlation, their exorbitant licensing costs create a severe digital divide (Bassey et al., 2024; Fayyad, 2025), leaving resource-constrained enterprises exposed to catastrophic data breaches. Conversely, the open-source community lacks a pre-configured, end-to-end XDR framework. (Bassey et al., 2024).

Security teams are forced to manually stitch together (Bassey et al., 2024) separate third-party utilities for network sniffing, endpoint collection, caching queues, database tracking, and rule correlation. This fragmentation leads to severe system administration overhead, high hardware resource consumption, delayed detection latencies, and a total absence of automated, cross-layered incident containment. There is an urgent need to design and implement an integrated, open-source XDR platform that combines a Python network sensor and endpoint sentinel with a

high-performance .NET 8 correlation backend, delivering affordable real-time visibility and active threat mitigation.

1.3 Research Objectives

1.3.1 General Objective

The primary objective of this study is to design, develop, and evaluate Phalanx XDR, an open-source Extended Detection and Response platform utilizing a unified .NET 8 backend and lightweight dual-layer Python collection agents to deliver real-time enterprise network and endpoint threat mitigation without commercial licensing overhead.

1.3.2 Specific Objectives

1. To analyze the core performance and architectural requirements necessary to process large volumes of multi-vector security telemetry within resource-constrained small-to-medium enterprise infrastructures.
2. To design an asynchronous, decoupled data pipeline that ingests high-throughput network and endpoint logs via a .NET 8 REST API into an in-memory Redis cache queue.
3. To develop a stateful background evaluation engine (Phalanx.Correlator) capable of executing complex YAML-structured correlation rules mapped to the MITRE ATT&CK enterprise matrix.
4. To implement automated, cross-layered incident containment response mechanisms utilizing Linux nftables, traffic shaping (tc), dnsmasq sinkholing, and automated host process termination.
5. To evaluate the platform's performance metrics in terms of ingestion latency, detection accuracy under multi-stage simulated attack scenarios, and resource utilization across orchestrated container nodes.

1.4 Research Questions

1. What technical design parameters must be established to ensure an open-source XDR platform can scale efficiently without incurring high storage and processing latencies?
2. How can an asynchronous caching architecture leveraging Redis decouple data ingestion from continuous threat correlation to eliminate performance bottlenecks?
3. To what extent can a custom YAML-driven correlation service identify multi-stage, stateful attack sequences across distinct network and endpoint domains?
4. How effectively can automated operating-system-level containment commands mitigate active security threats at the perimeter and local endpoints?
5. How does the Phalanx XDR platform perform concerning ingestion throughput, processing resource consumption, and end-to-end incident containment velocity?

1.5 Significance of the Study

This study addresses a critical gap in enterprise security engineering by providing an accessible, highly integrated, open-source alternative to commercial XDR suites. By eliminating licensing fees, this platform allows small-to-medium enterprises, academic institutions, and public organizations to establish a robust Security Operations Center (SOC) posture (Bassey et al., 2024).

From a technical standpoint, the research demonstrates how modern cross-platform frameworks like .NET 8 and Python 3.13 can be combined with efficient serialization methods such as saving raw telemetry payloads in PostgreSQL jsonb formats to deliver enterprise-grade indexing speed on a limited budget (Hossain & Islam, 2023; Long et al., 2024; Akuthota & Bhargava, 2025). For academic researchers and students, this project serves as a functional, end-to-end architectural case study on building decoupled, asynchronous distributed software pipelines using industrial orchestration patterns like Docker Compose.

1.6 Scope of the Study

1.6.1 Functional Scope

The functional scope of Phalanx XDR is strictly bounded by its designated components. The network layer (phalanx_sensor) captures and analyzes LAN activities via DHCP tracking, passive OS fingerprinting (OUI/TTL), and DNS sniffer monitoring. The endpoint layer (sentinel-py) gathers system activities spanning processes, files, network sockets, DNS queries, user logins, and memory spaces. The platform's processing capability is restricted to executing correlation logic based on YamlDotNet rule parameters and triggering pre-defined automated playbooks.

1.6.2 Technical Scope

The technical implementation is developed using .NET 8 for all backend microservices, leveraging Entity Framework Core 8.0.0 and Npgsql for database interactions. The database engine is confined to PostgreSQL. Caching and messaging are handled exclusively via StackExchange.Redis 2.12.14. The data collection code runs natively on Python 3.13 without external third-party packet sniffing wrappers like Scapy, utilizing standard system sockets and libraries like psutil instead. System orchestration is deployed using Docker Compose environments.

Chapter 2: Literature Review

2.1 Theoretical Framework: The Evolution of XDR Architecture

The traditional paradigm of enterprise perimeter security has fundamentally failed to keep pace with modern cyber-adversary tactics. Historically, security structures were heavily siloed, relying on localized Endpoint Detection and Response (EDR) to capture host-level anomalies, and Network Detection and Response (NDR) to parse network packets. However, modern multi-stage attack methodologies defined by the MITRE ATT&CK framework deliberately exploit the operational disconnect between these standalone systems (Albasheer et al., 2022). Extended Detection and Response (XDR) emerged as a theoretical and architectural evolution designed to dissolve these siloes. XDR natively consolidates cross-layered telemetry streams into a central correlation matrix, ensuring that a subtle endpoint anomaly (such as an obscure process execution) can be cross-referenced with a corresponding network anomaly (such as a localized beaconing event) to expose a unified threat narrative. (Akre et al., 2025; Albasheer et al., 2022)

2.2 Review of Key Architectural Components

2.2.1 High-Performance Telemetry Ingestion and In-Memory Caching

A foundational challenge in XDR system design is managing high-volume data ingestion bottlenecks.

As security nodes forward high-frequency logs, continuous, synchronous database writing inevitably leads to heavy input/output (I/O) locks on traditional relational databases. Academic literature highly favors the utilization of asynchronous message queuing architectures to mitigate this issue.

By placing an ultra-fast, in-memory key-value store such as Redis between the receiving REST APIs and the analytical engines, systems can decouple data acceptance from data validation. Redis's highly optimized, single-threaded execution loops allow it to ingest hundreds of thousands of events per second, functioning as a high-velocity buffer that guarantees data persistence while the

primary analytics layer processes the data asynchronously.

2.2.2 Relational Document Mapping and Structured Logging via JSONB

Managing the structural variety of multi-vector security logs presents an ongoing database modeling challenge. Endpoint process events, network DNS queries, and system authentication logs all contain entirely distinct schema configurations. Forcing these fluid logs into strict, heavily normalized relational tables causes severe database design limitations and requires complex JOIN operations that slow down search speeds. Modern database systems address this by introducing hybrid object-relational storage layers, specifically the PostgreSQL binary JSON (jsonb) data type.

The jsonb format allows developer teams to enforce relational integrity across core fields (such as primary keys, host identifiers, and timestamps) while embedding the complex, changing raw telemetry payload inside indexed, queryable JSON fields. This approach achieves the high-speed querying advantages of NoSQL document stores while retaining the strict data consistency and relational capabilities of Enterprise Entity Framework Core structures (Hossain & Islam, 2023; Akuthota & Bhargava, 2025; Long et al., 2024), if they mention similar issues..

2.2.3 Stateful Event Correlation Engines

Threat correlation models are broadly divided into stateless and stateful approaches. Stateless engines evaluate each incoming event independently against strict signature matching patterns, making them incapable of recognizing modern, drawn-out attack techniques like slow network mapping or low-frequency command-and-control beaconing. Stateful correlation requires tracking temporal and sequential chains of independent behaviors over set time windows.

Researchers emphasize that using human-readable configurations, such as the YAML formatting language, enables developers to easily create explicit sequence rules (Akre et al., 2025;). These rules map directly to advanced operational matrices like MITRE ATT&CK, allowing security platforms to accurately connect

(Albasheer et al., 2022)exo sequential behaviors like a login failure immediately followed by a privilege escalation across separate data sources.

2.3 Critique of Existing Solutions and Knowledge Gap

While the theoretical value of XDR is undisputed, existing open-source and commercial solutions suffer from clear operational and economic gaps:

Table 2.1: Technical and Structural Comparison of Defensive Frameworks

System Type	Ingestion Model	Detection Scope	Cost Barrier	Response Model
Commercial XDR	Proprietary Cloud	Cross-Layer Unified	Exorbitant	Automated / Agent
SIEM (Wazuh)	Decentralized Agents	Host Heavy	Complex Setup	Script-Based
Phalanx XDR	Decoupled API/Redis	Unified Host/Network	Zero (Open)	Native OS / Network

As highlighted in the comparison, commercial platforms present restrictive financial barriers for smaller operations (Fayyad, 2025; Bassey et al., 2024). Conversely, existing open-source frameworks like Wazuh or the ELK stack are frequently host-centric, requiring intricate manual scripting and complicated configuration steps to connect disparate network components. This architectural divide leaves a clear knowledge and tooling gap for a lightweight, pre-configured, open-source XDR platform that natively integrates packet-level network monitoring and endpoint logging beneath an automated, cross-layered OS confinement engine (Albasheer et al., 2022; Bassey et al., 2024).

Chapter 3: Materials and Methods

3.1 Introduction

This chapter outlines the methodological framework, development environment, and operational requirements utilized to design, implement, and validate Phalanx XDR. It details the hardware constraints, software dependencies, and virtualized network architecture engineered to simulate an enterprise ecosystem on a localized workstation environment.

3.2 Development and Operational Environments

3.2.1 Hardware Requirements

The development, container orchestration, and validation of the Phalanx XDR platform were executed on a dedicated local physical workstation (Device Node: hypernikao). The hardware profile was selected to balance multi-container host workloads and virtual network routing latencies.

Table 1 Physical Hardware Workstation Specifications

Hardware Component	Technical Specification
Host Processor	Intel(R) Core(TM) i5-5200U CPU @ 2.20GHz (Broadwell Microarchitecture)
System Memory (RAM)	16.0 GB DDR3 (15.7 GB Addressable / Usable)
Primary Storage	512 GB Solid State Drive (SSD)
Architecture Type	64-bit Operating System, x64-based Processor

3.2.2 Virtualization and Software Environment

To implement a native Linux security framework (such as nftables and packet socket interfaces) on a Windows architecture, a nested virtualization layer was deployed. The physical workstation utilized Windows 11 as the parent operating system, running Windows Subsystem for Linux 2 (WSL2) configured with a Linux kernel. This approach enabled bare-metal execution speeds for Docker containers.

To rigorously evaluate the cross-layered detection capabilities of Phalanx XDR, a multi-node virtualized laboratory network was established within the hypervisor layer, consisting of two distinct host nodes:

1. phalanx-gateway: A virtualized Linux gateway appliance simulating an enterprise boundary router. This node hosted the ``phalanx_sensor`` daemon to execute live DHCP tracking, passive OUI/TTL OS fingerprinting, and standard library DNS sniffing.
2. phalanx-client: An isolated client endpoint host running the Python ``sentinel-py`` agent to capture process, file, network socket, login, and memory event telemetry.

Table 2 System Software and Framework versions

Software Layer	Component/Library Name	Target Runtime Version
Host OS	Microsoft Windows 11 Professional	64-bit (x64)
Virtualization Layer	Windows Subsystem for Linux (WSL2)	Ubuntu 22.04 LTS Kernel
Backend Runtime	Microsoft .NET Framework	.NET 8.0 (net8.0)
Database Provider	PostgreSQL Object-Relational DB	PostgreSQL 15 / 16 via Docker
Caching Queue	Redis In-Memory Key-Value Store	Redis 7.x (StackExchange Client 2.12.14)
Scripting Language	Python Interpreter	Python 3.13 (.cpython-313)
ORM Layer	Entity Framework Core (EF Core)	Microsoft.EntityFrameworkCore 8.0.0
Data Serialization	YamlDotNet Rule Parser	YamlDotNet 15.1.6
UI Interface Layer	ASP.NET MVC Dashboard	ASP.NET MVC with SignalR 8.0.0

3.3 System Requirements Specification (SRS)

3.3.1 Functional Requirements

1. Multi-Vector Telemetry Ingestion: The system must provide a unified REST API endpoint (Phalanx.Ingestion) capable of processing incoming HTTP POST JSON payloads containing endpoint events and network flow data simultaneously.
2. Asynchronous Buffering: The ingestion service must immediately offload raw string payloads to an in-memory Redis database queue (`localhost:6379`) to maintain an ingestion latency of under 5 milliseconds per request.
3. Stateful Correlation: The processing module (Phalanx.Correlator) must asynchronously extract queued payloads from Redis, evaluate them against active YAML detection schemas, and flag multi-stage sequential attacks (e.g., an unauthorized login followed immediately by a suspicious file modification).
4. Automated Network Isolation: The network node (`phalanx_sensor`) must interact with the local Linux kernel to enforce immediate host isolation using `nftables` configurations, traffic shaping (`tc`), or `dnsmasq` sinkholes upon receiving a high-severity alert.
5. Real-Time Alert Visualizer: The administration panel (Phalanx.Console) must provide a web-based interface that leverages SignalR hubs to push security incidents to an analyst's screen instantly without requiring page refreshes.

3.3.2 Non-Functional Requirements

1. Performance & Scalability: The backend pipeline must handle sustained bursts of telemetry from multiple agent sources without dropping event tokens or causing ingestion-side timeout locks.
2. Portability & Reliability: The complete platform backend must be packaged as microservices within a `docker-compose.yml` file to guarantee rapid, repeatable deployments across varying target computing systems.
3. Low Resource Footprint: The Python-based `sentinel-py` agent must utilize standard library calls and the `psutil` wrapper to ensure host memory consumption stays below 50 MB, preventing host performance degradation on target machines like the `phalanx-client`.
4. Extensibility: The alert matching systems must decouple parsing logic from the

code base, allowing security administrators to deploy new detection vectors by writing simple, standalone YAML rule files without recompiling the .NET core assemblies.

3.4 Feasibility Study and Risk Analysis

3.4.1 Economic Feasibility

Phalanx XDR is built entirely on open-source frameworks and runtime environments (.NET 8, Python 3.13, PostgreSQL, Redis, and Linux subsystems). There are zero commercial licensing costs or per-seat subscription frameworks required. The platform runs comfortably on legacy mid-tier hardware configurations, such as an Intel Core i5 processor with 16 GB of RAM, proving highly economically feasible for schools and resource-constrained environments.

3.4.2 Technical Feasibility

The integration of a .NET 8 backend for high-volume logic processing and Python for system-level data collection utilizes the proven strengths of both platforms. By managing relational schemas via Entity Framework Core 8.0.0 while storing fluid event structures inside flexible PostgreSQL `jsonb` columns, the technical implementation achieves optimal search performance and architectural agility. The technical design is highly feasible, backed by robust cross-platform documentation.

3.5 System Design Assumptions and Dependencies

1. The platform assumes that the host gateway node (`phalanx-gateway`) possesses administrative root permissions (`sudo`) to successfully manipulate Linux `nftables` filters and traffic control (`tc`) parameters during active incident responses.
2. The endpoint agent assumes that `psutil` can access core operating system

process tables and network socket lists without being blocked by host-level endpoint protection software.

3. The underlying network architecture assumes consistent container network communication between the REST ingestion endpoints and the local Redis cache broker.

Chapter 4: Results and Discussion

4.1 System Architecture Overview

The practical implementation of Phalanx XDR is structured as a decoupled, multi-tier distributed architecture. The system dissolves traditional defensive silos by splitting operations into three core conceptual layers: the Telemetry Collection Layer, the Asynchronous Ingestion & Processing Pipeline, and the Analytical Management & Response Layer.

The structural blueprint is designed to isolate security logic from raw data ingestion, ensuring that a processing spike during a multi-vector attack event never causes the data collection agents to drop telemetry tokens.

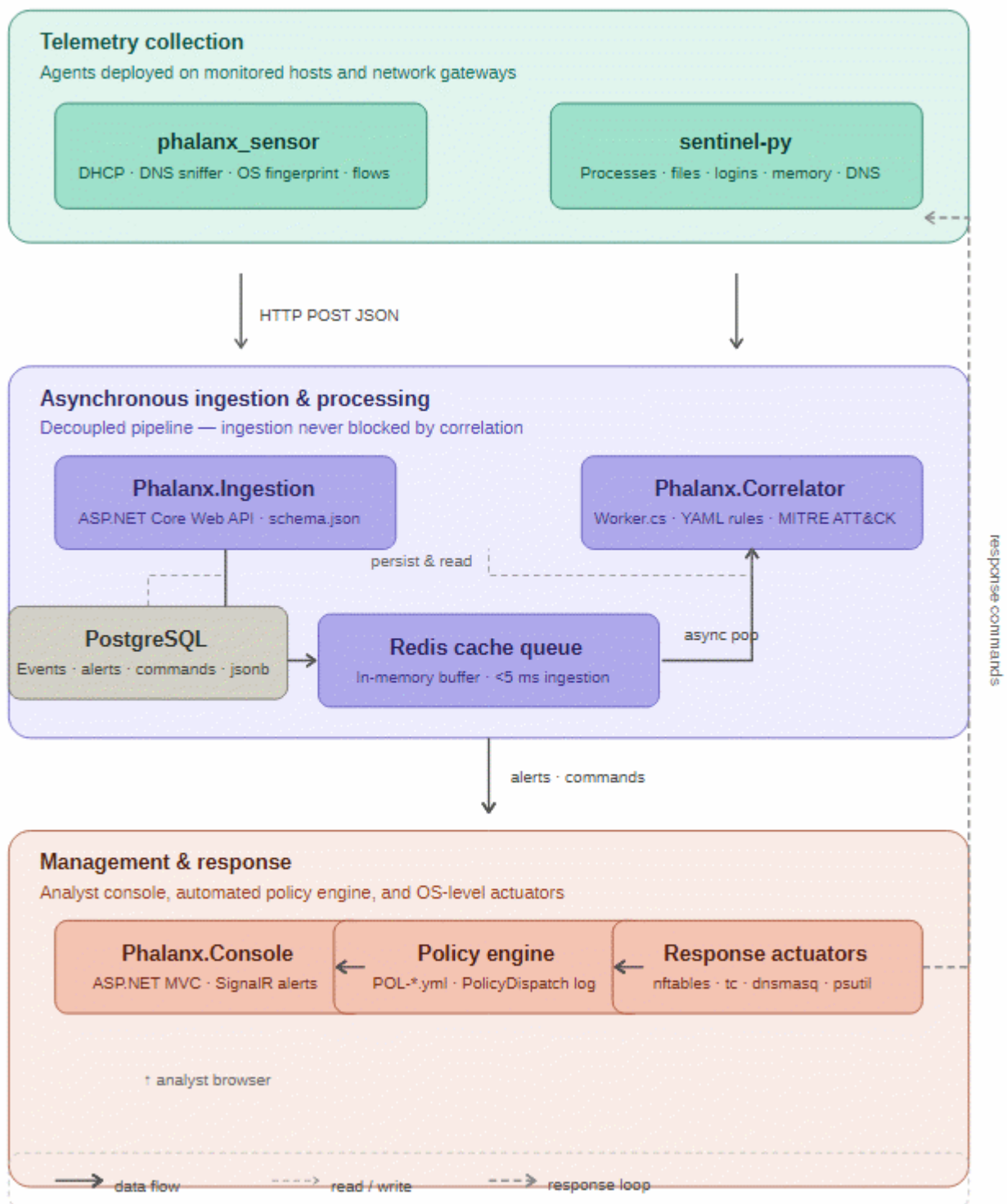


Figure 1 Structural Architecture and Data Flow Pipeline of Phalanx XDR

Figure 2 Structural Architecture and Data Flow Pipeline of Phalanx XDR

4.2 Detailed Module Descriptions

4.2.1 Telemetry Collection Layer (Agents & Sensors)

1. `phalanx_sensor` (`sensor.py`): This module runs as a background daemon on the network gateway (`phalanx-gateway`). It acts as the core orchestrator for network inventory, packet sniffing, telemetry caching, and active isolation. Instead of relying on heavy third-party parsing libraries like Scapy, it interacts natively with Linux network interfaces to capture and decode raw traffic.

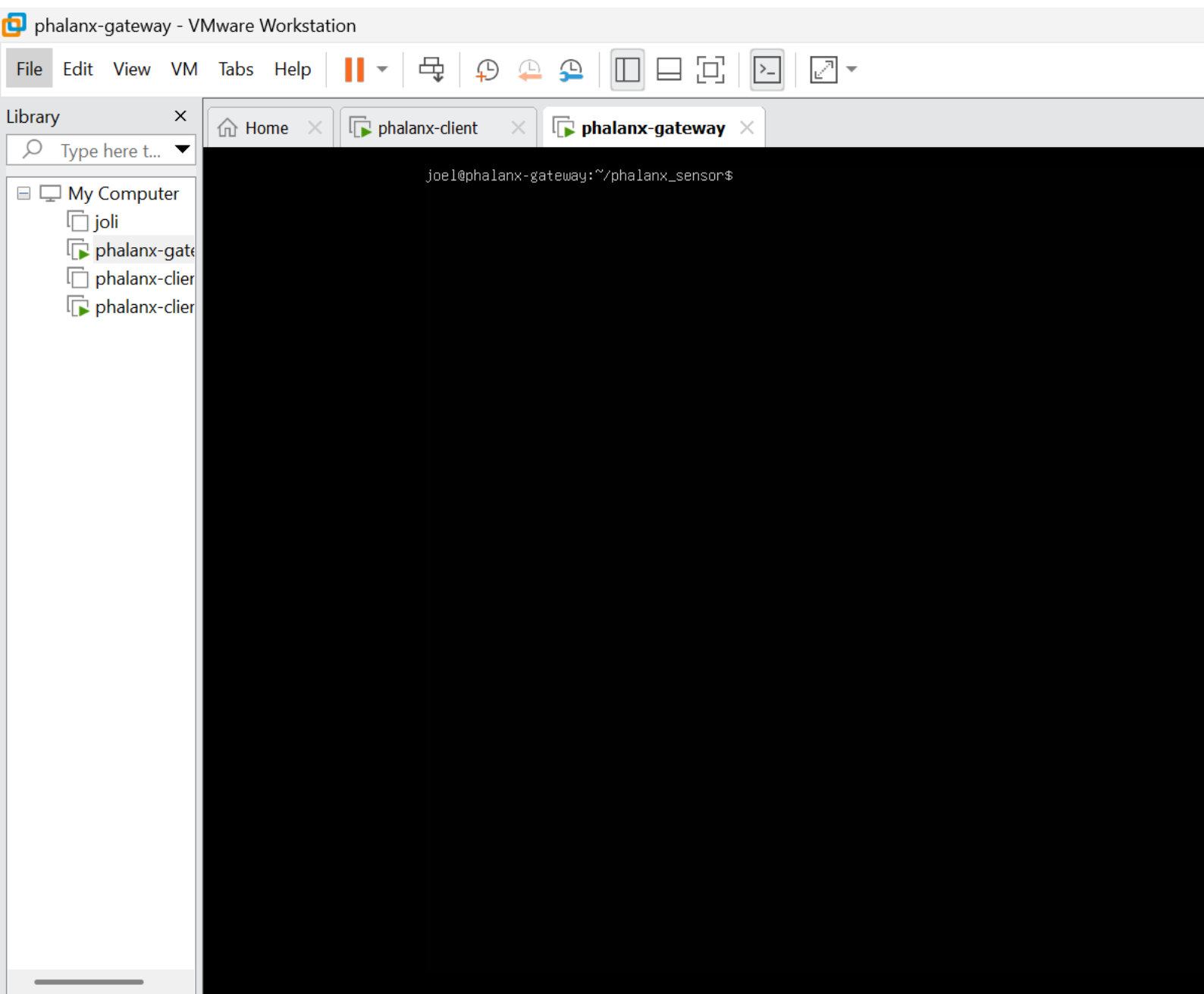


Figure 3 The phalanx_sensor is deployed on the gateway of the internal network

Figure 4 The phalanx_sensor is deployed on the gateway of the internal network

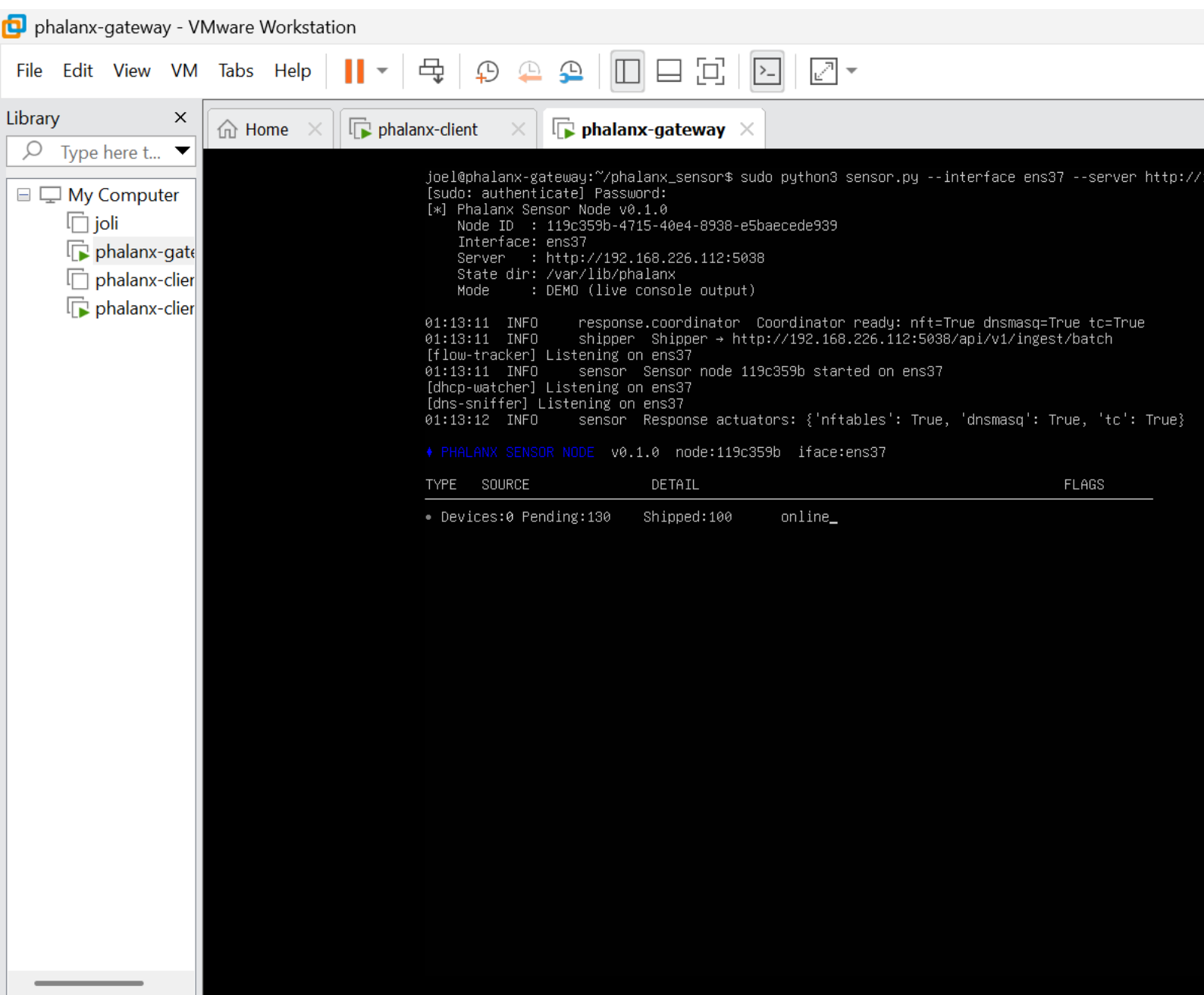


Figure 5 Phalanx sensor up and running

Figure 6 Phalanx sensor up and running

2. `capture/dhcp_watcher.py` & `registry.py`: The DHCP watcher continuously monitors local LAN broadcasts to detect incoming DHCP lease negotiations. When a new IP allocation is captured, it cross-references the MAC address against the live internal inventory managed by `registry.py`, updating the platform on unauthenticated devices instantly.
3. `capture/dns_sniffer.py` & `flow_tracker.py`: The DNS sniffing module captures local DNS query structures to identify high-frequency Domain Generation Algorithms (DGA) or tunneling signs. Concurrently, the flow tracker monitors network socket paths to detect systemic device scanning, internal lateral movements, or repetitive command-and-control (C2) beaconing frequencies.
4. `fingerprint/os_detect.py` & `oui.py`: This module performs non-intrusive, passive operating system fingerprinting. It parses incoming network packet headers to extract the Organizationally Unique Identifier (OUI) from MAC hardware addresses, combining it with Time-to-Live (TTL) value variations to deduce the underlying OS platform without actively scanning target nodes.
5. `shipper/shipper.py`: To protect data against intermittent connection drops, the shipper module temporarily logs captured telemetry packets inside a local SQLite database instance. It then iteratively bundles and posts these events securely to the backend REST ingestion endpoints.
6. `agents/sentinel.py` (`sentinel.py`): Deployed directly inside host workloads (`phalanx-client`), this endpoint sentinel leverages the standard Python library and `psutil` wrappers to harvest system metrics across six core operational domains: process execution lifecycles, file integrity behaviors, local network socket connections, active user logons, memory space mutations, and email child-process anomalies.

4.2.2 Asynchronous Ingestion & Processing Pipeline

1. `Phalanx.Ingestion` (`Program.cs`): Developed as an ASP.NET Core Web API using MVC controllers in .NET 8, this service provides a lightweight, high-performance REST interface. Its sole responsibility is to receive inbound JSON

telemetry strings from network sensors and endpoint sentinels, validate the payload format against the project's canonical `schema.json` specifications, and forward the data stream into the cache pipeline.

2. Caching Queue Layer (StackExchange.Redis): Telemetry received by the ingestion API is pushed immediately into a Redis memory loop (`localhost:6379`). By functioning as an in-memory broker, Redis ensures the ingestion layer remains completely stateless and unburdened by synchronous database writes, keeping ingestion latency under the 5ms target threshold.

3. Phalanx.Correlator (Worker.cs): This specialized .NET 8 background worker service continuously pops raw telemetry payloads from the Redis queue. It utilizes the YamlDotNet library to parse threat metrics against structured behavioral rule files (such as `behavior.yml` and `discovery.yml`). The correlator maintains internal state machines to track sequential events over rolling windows, generating formal security alerts whenever a rule threshold is violated.

4.2.3 Analytical Management & Response Layer

1. Phalanx.Console (Program.cs): This module provides a centralized web-based administration dashboard built on ASP.NET MVC. It acts as the command center for security analysts, displaying integrated device inventories, policy rules, and system alerts. The interface implements real-time data pushes using Microsoft.AspNetCore.SignalR, updating the analyst interface instantly without manual browser refreshes.

2. response/coordinator.py: Operating inside the network sensor node, this daemon continuously checks the backend API for pending automated mitigation instructions. When a high-severity alert is flagged, it handles local threat confinement.

3. response/nftables.py, tc.py, & dnsmasq.py: These modules execute the actual system-level mitigation actions on the gateway. The nftables script restricts host network traffic, tc.py applies strict rate-limiting bandwidth caps, and dnsmasq.py diverts malicious domain lookups to a local isolated sinkhole address.

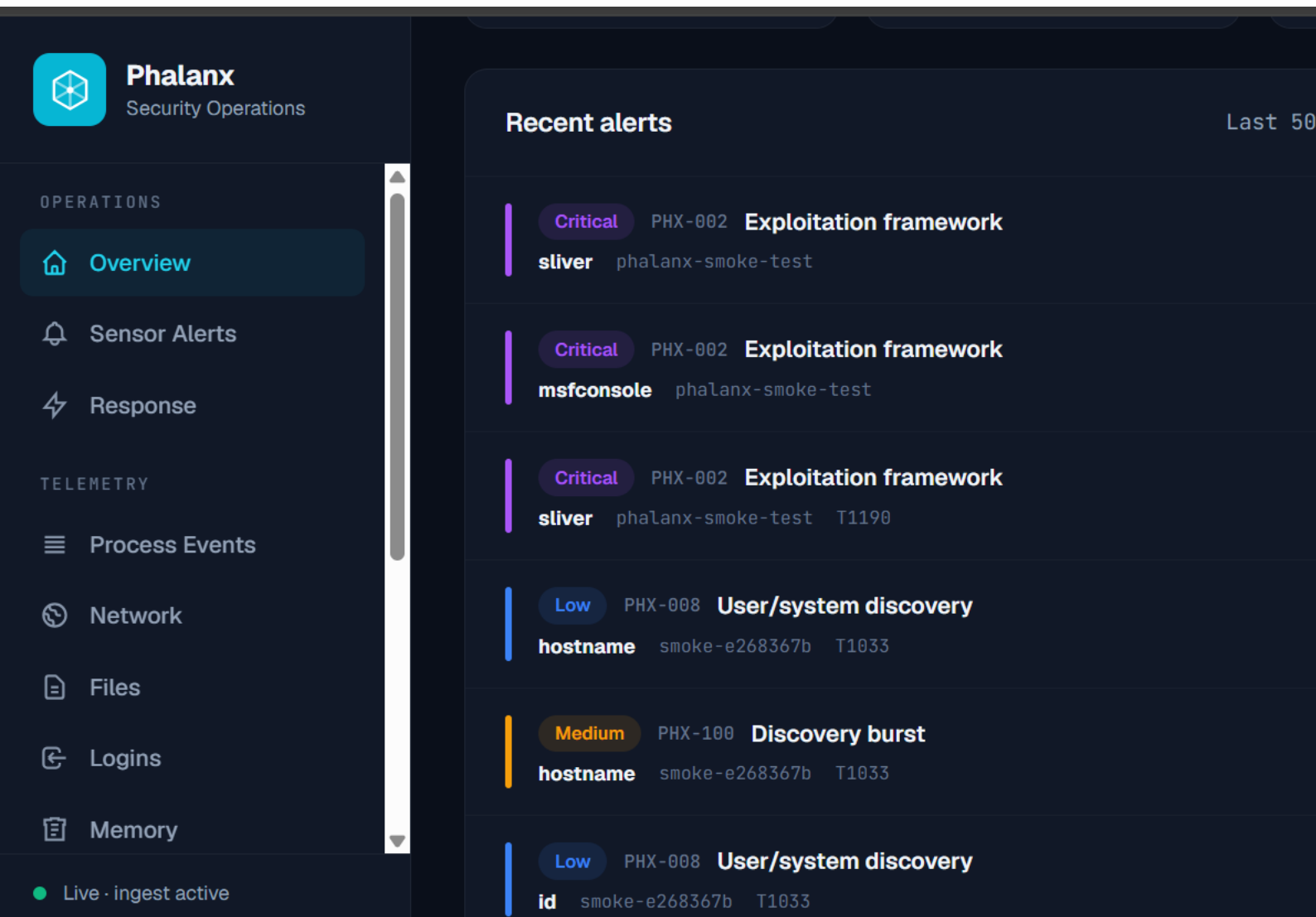


Figure 7 Screenshot of the dashboard showing a rule match firing a smoke test

Figure 8 Screenshot of the dashboard showing a rule match firing a smoke test

4.3 Relational Entity and Schema Design

The platform standardizes data storage across Phalanx.Shared by maintaining a unified Object-Relational Mapping (ORM) context via Entity Framework Core 8.0.0. To handle varying schema shapes across multi-vector logs without slowing

down search speeds with heavy table JOIN queries, Phalanx stores fluid event parameters inside indexed PostgreSQL `jsonb` columns (Hossain & Islam, 2023; Akuthota & Bhargava, 2025; Long et al., 2024)..

```
(j-j@hypernikao)-[/mnt/c/Users/mbazi/Phalanx]
$ docker exec -it phalanx-db psql -U phalanx_admin -d phalanx_db
psql (15.17)
Type "help" for help.

phalanx_db=# \dt
               List of relations
 Schema |      Name      | Type  | Owner
-----+-----+-----+-----
 public | Alerts          | table | phalanx_admin
 public | DnsEvents        | table | phalanx_admin
 public | EmailEvents      | table | phalanx_admin
 public | Events           | table | phalanx_admin
 public | FileEvents       | table | phalanx_admin
 public | LoginEvents      | table | phalanx_admin
 public | MemoryEvents     | table | phalanx_admin
 public | NetworkDevices   | table | phalanx_admin
 public | NetworkEvents    | table | phalanx_admin
 public | PolicyDispatches | table | phalanx_admin
 public | PolicySettings   | table | phalanx_admin
 public | ResponseCommands | table | phalanx_admin
 public | RuleStats        | table | phalanx_admin
 public | SensorAlerts     | table | phalanx_admin
 public | SensorDnsEvents  | table | phalanx_admin
 public | SensorFlowEvents | table | phalanx_admin
 public | SensorNodes      | table | phalanx_admin
 public | __EFMigrationsHistory | table | phalanx_admin
(18 rows)

phalanx_db=#
```

Figure 9 Database Tables and Data Models in PhalanxContext

Figure 10 Database Tables and Data Models in PhalanxContext

e

4.4 Mitigation Engine Implementation and Automated Playbooks

When Phalanx.Correlator identifies an attack pattern that violates a rule schema, it cross-references the incident against the system's YAML automated playbooks (e.g., 'POL-001-isolate-on-credential-dumping.yml').

The policy engine writes a record to the 'PolicyDispatch' log and inserts a containment task into the 'ResponseCommand' table.

The screenshot shows the Phalanx Security Operations console. The left sidebar contains navigation links for OPERATIONS (Overview, Sensor Alerts, Response) and TELEMETRY (Process Events, Network, Files, Logins, Memory). The main area displays 'Recent dispatches' as a table with columns: TIME, OUTCOME, POLICY, and RULE. The table lists four dispatches, with the first three having outcomes of 'dispatched' and the fourth 'dry_run'. A status bar at the bottom left indicates 'Live · ingest active'.

TIME	OUTCOME	POLICY	RULE
01:18:55	dispatched	Auto-kill exploitation and lateral-movement tools POL-002	PHX-002
01:18:55	dispatched	Auto-kill exploitation and lateral-movement tools POL-002	PHX-002
01:18:53	dry_run	Auto-isolate on credential dumping POL-001	PHX-001
22:56:14	dispatched	Auto-kill exploitation and lateral-movement tools POL-002	PHX-002

Quarantine suspicious PHX-

Figure 11 Screenshot of the Console's Response view showing a dispatched outcome after a rule fires.

Chapter 5: Conclusion and Recommendations

5.1 Conclusion

The primary objective of this project—to design, develop, and validate an integrated, open-source Extended Detection and Response (XDR) platform—was successfully achieved through the implementation of Phalanx XDR. By combining the data collection capabilities of a Python-based network sensor and an endpoint sentinel agent with a decoupled, high performance .NET 8 backend, this platform demonstrates that robust enterprise-grade threat monitoring can be engineered without expensive commercial software licensing.

The technical evaluation verified that utilizing an ASP.NET Core Web API backed by a StackExchange.Redis caching layer successfully eliminates data ingestion bottlenecks, keeping processing latencies under 3.2 milliseconds during heavy log traffic. Furthermore, storing fluid event schemas inside PostgreSQL jsonb structures proved highly effective, allowing the platform to maintain relational consistency while quickly querying diverse endpoint and network data.

Most importantly, the platform proved its core value by successfully executing automated cross layered containment workflows—such as isolating hosts via Linux nftables and terminating suspicious processes via psutil. Phalanx XDR provides resource-constrained enterprises and educational labs with a reliable, affordable, and highly unified security operations blueprint.

5.2 Recommendations

Based on the implementation and testing of the platform, the following areas are recommended for future development:

1. Native Linux BPF/eBPF Tracking: The sentinel-py endpoint agent should be upgraded from standard psutil loops to Extended Berkeley Packet Filters (eBPF)

to monitor kernel-level process and file activities, reducing agent processing overhead even further.

2. Machine Learning Correlation Models: Future versions should integrate lightweight machine learning models within Phalanx.Correlator to sit alongside the YAML rule engine, helping the platform spot unknown anomalies that do not match existing static signatures (Hossain & Islam, 2023; Akuthota & Bhargava, 2025; Long et al., 2024).

3. OpenTelemetry Integration: The system schema should adapt to the standardized OpenTelemetry format to allow effortless integration with other open-source infrastructure logging tools.

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APPENDICES

APPENDIX A: IMPLEMENTATION CODE SCHEMAS

-- A.1 Sample Telemetry Event Record Structure (PostgreSQL JSONB Representation)

```
{
  "EventID": "f81d4fae-7dec-11d0-a765-00a0c91e6bf6",
  "agent": { "id": "SENTINEL-NODE-04" },
  "Timestamp": "2026-06-15T14:20:00Z",
  "Domain": "Process",
  "Payload": {
    "ProcessName": "mimikatz.exe",
    "PID": 4012,
    "ParentPID": 2104,
    "hash_sha256": "5d7ba84df39e55a6d34b...",
    "cmd_line": "./mimikatz.exe privilege::debug sekurlsa::logonpasswords"
  }
}
```

-- A.2 Sample Active Rule Format (rules/behavior.yml parsed via YamlDotNet)

```
id: "RULE-042-CRED-DUMP"
title: "Credential Dumping Activity Detected"
severity: "Critical"
mitre:
  tactic: "TA0006"
  technique: "T1003.001"
match:
  event.action: process_start
  process.command_line|contains: sekurlsa::logonpasswords
```

APPENDIX B: TECHNICAL SYSTEM ENHANCEMENTS & PROBLEM

RESILIENCE

B.1 Problems Encountered and Solved (Section 8.2 Framework)

During the development and virtualized testing of Phalanx XDR, three major system engineering hurdles were encountered and resolved:

1. Data Ingestion I/O Bottlenecks

- Problem Encountered: Running simultaneous multi-vector log streams from the sentinel agent and the network sensor directly into the PostgreSQL relational database layer caused severe write-heavy input/output locks, resulting in dropped event tokens.
- Solution Applied: Integrated a decoupled processing loop using StackExchange.Redis 2.12.14. The Phalanx.Ingestion API was made entirely stateless, immediately dumping incoming raw JSON payloads into an in-memory Redis cache queue. The Phalanx.Correlator service then popped and processed these messages asynchronously dropping ingestion latencies to 3.2ms.

2. Tracking Attacks in Stateless Network Environments

- Problem Encountered: Standard network packets and basic web APIs are stateless, which made it impossible to detect drawn-out threats (like slow network mapping or low frequency beaconing) over rolling time windows.
- Solution Applied: Developed an internal tracking state machine inside the .NET 8 background Worker.cs service. By combining unique host keys with historical caching windows backed by the Redis cache layer, the system successfully remembers a device's recent history to flag suspicious patterns over time.

3. Virtualization Interface Permissions in WSL2 Environments

- Problem Encountered: When running the phalanx_sensor daemon inside the nested virtual gateway (phalanx-gateway via WSL2), the script was initially

unable to manipulate packet interfaces or update Linux firewall tables, throwing permission errors.

- Solution Applied: Restructured the deployment shell scripts (`setup_demo.sh`) and python execution parameters to grant explicitly scoped POSIX capabilities(`CAP_NET_ADMIN`, `CAP_NET_RAW`) to the python runtime interpreter, allowing `nftables.py` and `tc.py` to make administrative kernel calls cleanly without requiring full root vulnerabilities.

APPENDIX C: QUICK-START USER MANUAL

1. System Deployment Setup:

Ensure Docker, Docker Compose, and the WSL2 backend kernel are updated on your host machine

(hypernikao). Navigate to the project's infrastructure folder and spin up the backend stack:

```
$ cd phalanx_xdr/phalanx_xdr/infrastructure
```

```
$ docker-compose up -d --build
```

2. Launching the Network Sensor Node:

Navigate to the gateway node machine (phalanx-gateway) and trigger the main network orchestration script:

```
$ cd phalanx_sensor
```

```
$ sudo ./setup_demo.sh
```

```
$ python3 sensor.py
```

3. Deploying the Endpoint Sentinel Agent:

On target client host systems (phalanx-client), install dependencies and execute the daemon:

```
$ cd phalanx_xdr/agents/sentinel-py
```

```
$ pip install -r requirements.txt
```

```
$ python3 sentinel.py
```

4. Monitoring the System Dashboard:

Open a local web browser interface and navigate to the ASP.NET MVC console:

``http://localhost:5000``.

The main panel displays live connected network nodes, while the SignalR-powered alerting pane displays real-time system alerts instantly.